SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: *Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 50%

Duration: 1 Hour

QUESTIONS: 5

Total Marks:50

Annexure A

Scenario Based Assignment

Code of Conduct:

*I **Narappagari Sravya(192472192)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:*sravya*

Annexure A

Scenario Based Assignment

1. **Scenario** : A government deploys AI-powered drones to deliver emergency medicines in a flood-affected region. The environment is highly dynamic—roads may suddenly become inaccessible, and weather conditions affect travel cost. The drone has access to: A heuristic estimate (straight-line distance), Partial real-time updates about blocked paths and Variable movement costs depending on terrain and weather. However, due to urgency, the drone must quickly decide routes with minimum delay and risk.

   Tasks:

i. Explain how Greedy Best-First Search would behave in this scenario and discuss its strengths and weaknesses under uncertainty

ii. Explain how A* would handle the same situation, including how it balances cost and heuristic

iii. Analyze the impact of heuristic accuracy on both algorithms

iv. Discuss how dynamic changes (blocked paths) affect search performance

v. Recommend the most suitable algorithm and justify your decision considering realtime constraints, optimality, and safety

2. **Scenario**: A smart city implements an AI system to optimize traffic signal timings across hundreds of intersections. The objective is to minimize Total waiting time, Fuel consumption and Traffic congestion. The system must continuously adjust signals based on traffic flow. However the search space is extremely large, traffic patterns change dynamically and the system may get stuck in locally optimal solutions

   Tasks. Formulate this as an optimization problem and explain why traditional search is inefficient

i. Explain how Hill Climbing would work in this scenario and identify its limitations

ii. Discuss issues like local maxima, plateaus, and ridges with real-world interpretation

iii. Explain how Simulated Annealing or Genetic Algorithms can overcome these limitations

iv. Recommend the best approach and justify your choice based on scalability and adaptability

3. **Scenario:** An autonomous rover is deployed on Mars to explore unknown terrain. It must: navigate safely to collect samples, avoid hazards (craters, rocks), operate with limited sensing capability and make decisions without a complete map. The rover updates its knowledge as it explores, making this a real-time decision-making problem.

   Tasks:

i. Explain why this problem requires an online search agent instead of offline search

ii. Analyze the challenges of partial observability and dynamic environments

iii. Describe how the rover builds and updates its internal model

iv. Compare online search with uninformed and informed search strategies

v.Justify the most suitable approach considering uncertainty, adaptability, and safety

4. **Scenario:** A university is designing an AI system to automatically generate exam timetables. The system must satisfy multiple constraints: No student should have overlapping exams, Limited number of exam halls, Certain subjects must be scheduled before others, Faculty availability constraints, Special accommodations for some students, The complexity increases as the number of courses and students grows.

   Tasks :

   Formulate the problem as a Constraint Satisfaction Problem (CSP)

i. Clearly define variables, domains, and constraints with explanation

ii. ii.Explain how backtracking search works in this scenario

iii. Discuss how constraint propagation (e.g., forward checking) improves efficiency

iv. Analyze real-world challenges and justify the best strategy for scalability

5. **Scenario:** Strategic Game AI for Real-Time Decision Making (Minimax & Alpha-Beta

Pruning) A gaming company is developing an AI opponent for a real-time strategy game. The AI must complete against human players, Make decisions within strict time limits, Evaluate complex game states with multiple possible moves, The decision tree is extremely large, making computation expensive.

Tasks:

i. Explain how the Minimax algorithm models this problem

ii. Analyze the computational challenges of large game trees

iii. Explain how Alpha-Beta pruning improves efficiency with detailed reasoning

iv. Design an evaluation function and justify the factors considered

v. Recommend a strategy for balancing optimality and real-time performance

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes wellreasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

1. AI-Powered Drone for Emergency Medicine Delivery

i. Greedy Best-First Search (GBFS)

Greedy Best-First Search selects the path that appears closest to the destination using only the heuristic value **h(n)** (straight-line distance).

Working

- Uses only the heuristic estimate.
- Expands the node that seems nearest to the goal.
- Ignores the actual travel cost.

Strengths

- Very fast.
- Requires less computation.
- Suitable for urgent situations.

Weaknesses

- May choose unsafe or blocked routes.
- Does not guarantee the shortest path.
- Performs poorly when the heuristic is inaccurate.

ii. A* Search

A* considers both the actual cost and estimated remaining cost.

Formula

f(n) = g(n) + h(n)

where

- **g(n)** = actual cost from the start
- **h(n)** = estimated cost to the goal

Working

- Continuously evaluates total path cost.

- Selects the route with the minimum overall cost.
- Can adapt when movement costs change.

Advantages

- Finds the optimal path.
- Avoids expensive or risky routes.
- More reliable than Greedy Search.

Disadvantages

- Requires more computation.
- Needs additional memory.

iii. Impact of Heuristic Accuracy

Greedy Best-First Search

- Accurate heuristic → Fast and good path.
- Poor heuristic → Wrong decisions and longer routes.

A*

- Accurate heuristic → Faster search and optimal solution.
- Less accurate heuristic → Still finds the optimal solution but explores more nodes.

iv. Effect of Dynamic Changes

When roads become blocked,

- Greedy Search may continue toward blocked paths before changing direction.
- A* recalculates the route considering updated movement costs.
- Frequent updates increase computation time for both algorithms.

v. Recommendation

Recommended Algorithm: A*

Justification

- Produces optimal routes.
- Considers both travel cost and distance.
- Adapts to changing environments.
- Provides safer navigation during emergencies.

2. Smart City Traffic Signal Optimization

Problem Formulation

Objective:

- Minimize waiting time
- Minimize fuel consumption
- Reduce traffic congestion

Variables:

- Green signal duration
- Signal sequence
- Signal timing

Traditional search is inefficient because:

- Search space is extremely large.
- Traffic changes continuously.
- Exhaustive search requires too much time.

i. Hill Climbing

Hill Climbing improves the current traffic signal timing step by step.

Advantages

- Fast.
- Easy to implement.
- Suitable for local optimization.

Limitations

- Can get stuck in local optimum.
- Cannot always find the best global solution.
- Sensitive to the starting solution.

ii. Local Maxima, Plateaus and Ridges

Local Maximum

A traffic signal timing appears best locally but better solutions exist elsewhere.

Plateau

Many signal timings produce nearly identical performance, making improvement difficult.

Ridge

The best solution requires several coordinated timing changes instead of one small adjustment.

iii. Simulated Annealing and Genetic Algorithm

Simulated Annealing

- Occasionally accepts worse solutions.
- Escapes local maxima.
- Gradually converges toward better solutions.

Genetic Algorithm

- Maintains multiple candidate solutions.
- Uses selection, crossover and mutation.
- Efficiently explores large search spaces.

iv. Recommendation

Recommended Approach: Genetic Algorithm

Reason

- Highly scalable.
- Handles dynamic traffic efficiently.
- Produces near-optimal solutions.
- Suitable for large smart city networks.

3. Autonomous Mars Rover

i. Why Online Search?

The rover has no complete map.

It must:

- Sense surroundings.
- Make decisions immediately.
- Learn while moving.

Offline search requires complete environmental knowledge, which is unavailable.

ii. Challenges

Partial Observability

- Limited sensor range.
    - Unknown obstacles.

Dynamic Environment

- Dust storms.
- Moving rocks.
- Unknown terrain.

These conditions increase uncertainty.

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iii. Internal Model

The rover continuously updates its internal map by

- Collecting sensor data.
- Identifying safe and unsafe regions.
- Recording explored paths.
- Updating future navigation plans.

iv. Online vs Uninformed vs Informed Search

Online Search	Uninformed Search	Informed Search
Learns while exploring	No heuristic	Uses heuristic
Suitable for unknown environments	Requires full exploration	Needs good heuristic
Adapts to changes	Less efficient	Faster than uninformed

v. Recommendation

Online Search Agent

Reason:

- Works without complete knowledge.
- Adapts to new obstacles.
- Improves safety.
- Supports real-time exploration.

4. University Exam Timetable (CSP)

CSP Formulation

Variables

Each course examination.

Example:

- Math
- Physics
- AI

Domains

Possible

- Dates
- Time slots
- Examination halls

Constraints

- No overlapping exams for students.
- Hall capacity should not exceed the limit.
- Faculty must be available.
- Some exams must occur before others.
- Special accommodation students require suitable halls.

ii. Backtracking Search

Backtracking assigns exams one by one.

If a constraint is violated:

- Undo the last assignment.
- Try another time slot.
- Continue until all exams are scheduled.

iii. Forward Checking

Forward checking removes invalid future assignments after each decision.

Benefits:

- Detects conflicts early.
- Reduces unnecessary search.
- Improves efficiency.

iv. Real-World Challenges

- Thousands of students.

- Hundreds of courses.
- Limited halls.
- Frequent timetable changes.

Best Strategy

Backtracking + Forward Checking + Constraint Propagation

This combination improves scalability and reduces computation time.

5. Strategic Game AI

i. Minimax Algorithm

Minimax assumes

- AI tries to maximize its score.
- Opponent tries to minimize the AI's score.

The algorithm evaluates future game states and selects the best move.

ii. Computational Challenges

Large games have

- Huge branching factor.
- Deep decision trees.
- High memory usage.
- Slow computation.

Real-time games cannot evaluate every possible move.

iii. Alpha-Beta Pruning

Alpha-Beta Pruning removes branches that cannot improve the final decision.

Benefits:

- Reduces the number of evaluated nodes.
- Produces the same optimal move as Minimax.
- Significantly speeds up decision making.

iv. Evaluation Function

The evaluation function may include

- Number of units.
- Remaining health.
- Resource availability.

- Territory control.
- Distance to objectives.
- Defensive strength.

Higher evaluation score means a stronger game position.

v. Recommendation

Recommended Strategy

Alpha-Beta Pruning with Minimax

Justification

- Produces high-quality decisions.
- Reduces computation.
- Meets real-time constraints.
- Maintains near-optimal performance in complex games.